# Data Structures and Algorithms · 21CSC201J

Unit 3 · Session 6 · SLO 1 — Solved Worksheet

**Name:** DEEP SAND DEEP SAND | **Reg. No:** RA2511026010557

## Unit â€“ 3 : Session â€“ 6 : SLO â€“ 1 | Stack Applications â€” Recursion

### I. Fill in the Blanks

• In recursion, each function call is stored in the STACK (call stack).
• The LIFO principle is followed by stacks.
• Stack is used internally to implement RECURSIVE function calls.
• Every recursive function must have a BASE condition to avoid infinite calls.
• The process of returning from a recursive function happens in the REVERSE order of calls.
• Each recursive call adds a new ACTIVATION RECORD (stack frame) to the stack.
• FACTORIAL (Tower of Hanoi / Fibonacci) is a classic example of a problem solved using recursion and stacks.
• The return address and local variables are stored in the function's ACTIVATION RECORD (stack frame).
• The system stack helps in managing NESTED (multiple) function calls during recursion.
• When the base case is reached, the stack starts to UNWIND (pop frames).